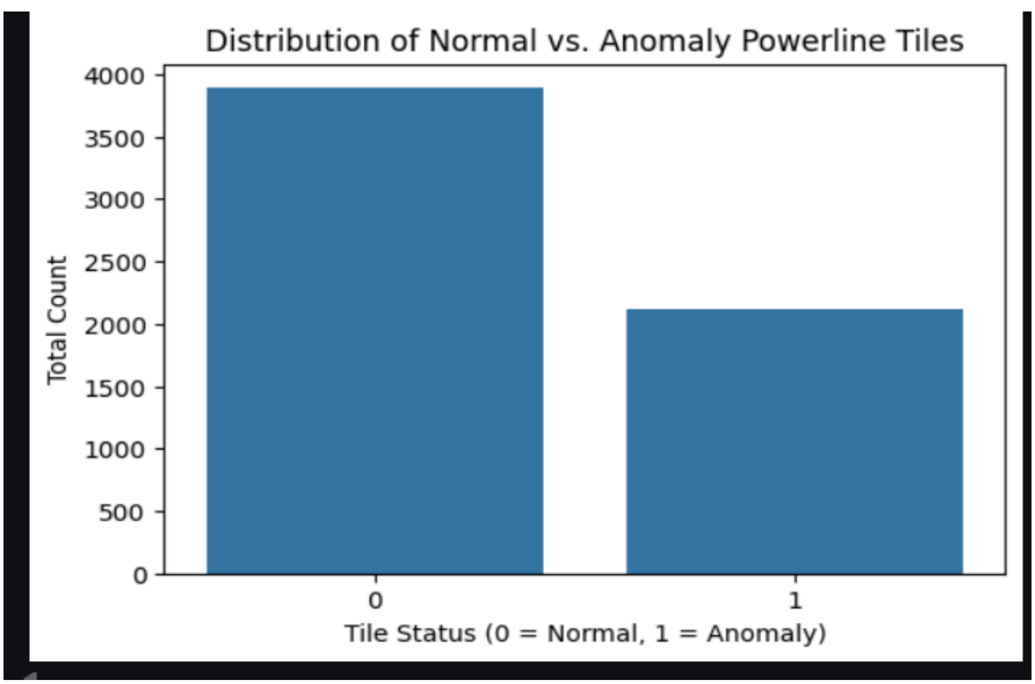
AI Based Thermal Powerline Hotspot Capstone Project

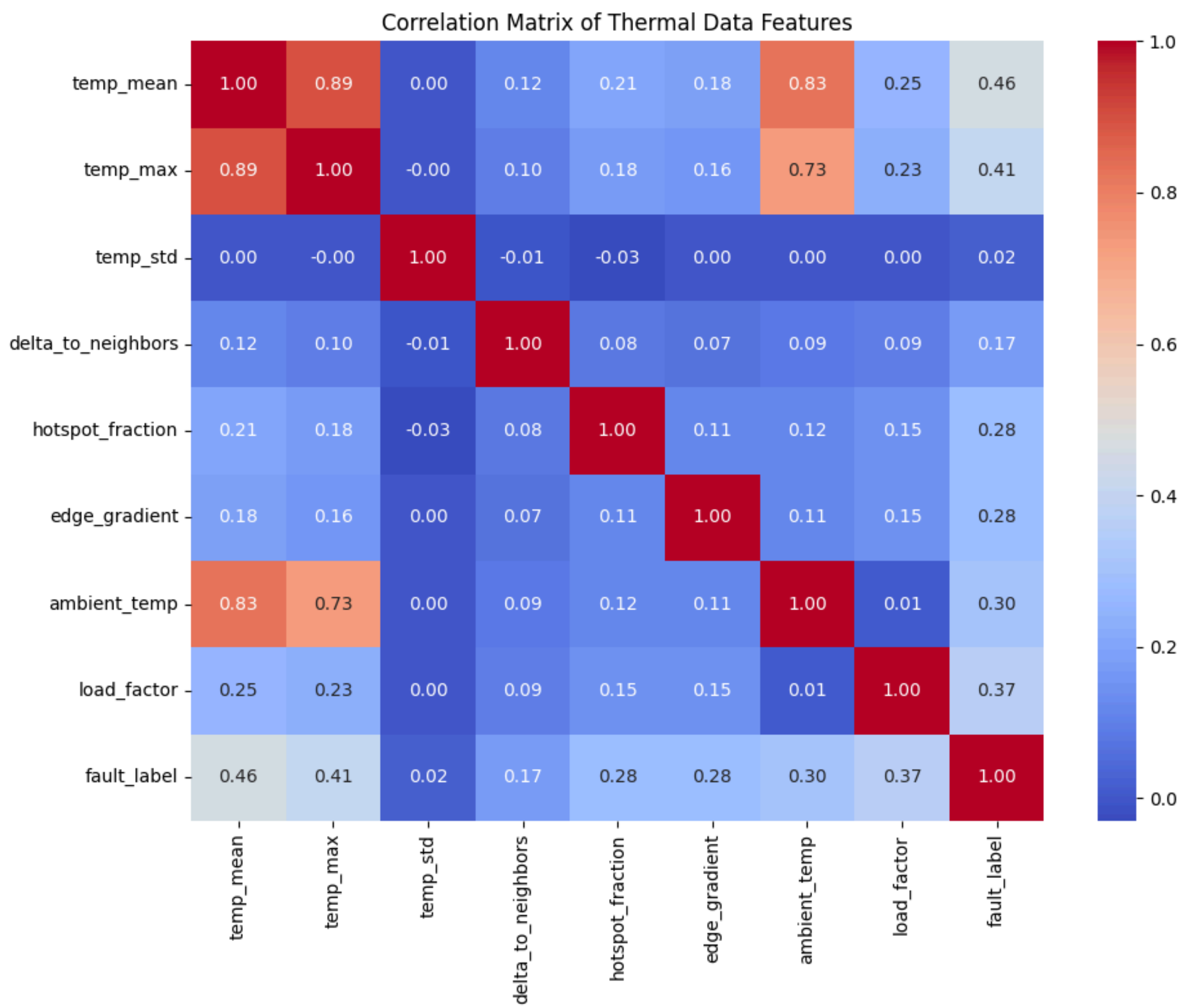
In this capstone,I will design an end-to-end AI pipeline to detect thermal hotspots in power lines and transmission towers using drone-based thermal inspection data.

Im gonna use this dataset which has the following features -
`temp_mean', 'temp_max', 'temp_std', 'delta_to_neighbors','hotspot_fraction', 'edge_gradient', 'ambient_temp', 'load_factor','fault_label'`

Next I am going to check for any missing data , upon further analysis , I see that there is no missing data in any of the columns.

Using Matplotlib and Seaborn to check for the number of abnormalities / fault_label in the data
And also generating a heatmap to understand the correlation between each feature
Warm ie red represents that both features rise together , and the Cold ie blue represents that when one feature rises the other one falls.





So lets now analyse
1. I see that the abnormalities around 2000 data points and the normal datapoints tend towards the 4000 mark
2. By the rule of thumb , i think that all the mapping above 0.7 or below -0.7 should be considered highly correlated by this rule , I see that
Temp_max , temp_mean_ temp_ambient r all correlated over 0.7
This also makes sense from a logical standpoint as if the outside weather ie ambient temperature is hot m the overall temperatures will naturally scale with it.
3. I see that none of the features have a strong correlation with the target_feature ie fault_label , the highest being temp_mean at 0.46b and temp_max at 0.41 , load_factor at 0.37 and both hotspot_fraction and edge_gradient at 0.28 are slightly related. This is due to the fact that there is no single cause of this fault .

Now lets move towards actually making the machine learning model ,
For that

1. Lets drop the target_variable

2. Lets assume spatial coordinates , im assuming it to be 100X60 as we have 6000 data points
3. Making features spatial_x and spatial_y

4. Using train test split on the dataset and also scaling using the standard scaling

Now im gonna use an MLP classifier and a Random forest model to compare the accuracy of the results .
MLP is a multi layer perceptron ie a multi layered neural network , whereas Random Forest is an ensemble tree-based classification mode .

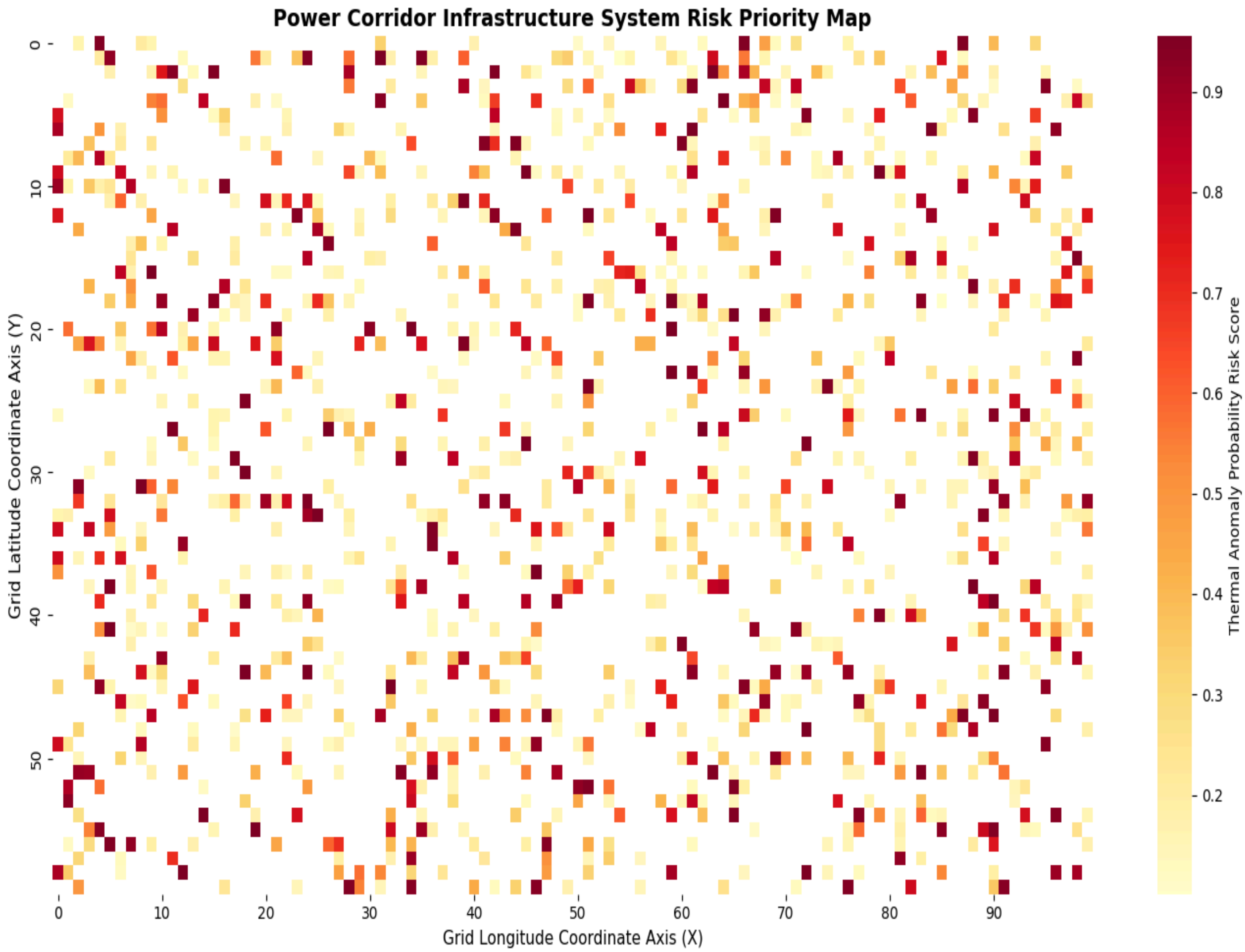
Lets first work on the RandomForestClassifier
For this i first ran Gridsearchcv to perform cross validation for the best parameters, which came out to be -`{'max_depth': 10, 'max_leaf_nodes': 75}`
Now im gonna use the RandomForestClassifier with those parameters to predict output on the test set we created using TrainTestSplit to further test the accuracy of those results using the metrics - Recall , Precision , f1 score

Here are the results ,
The confusion matrix -
```
[[671 106]
 [118 305]]
```
Precision- 0.74 , Recall - 0.72, f1 score -0.73
ROC-AUC score is 0.8537
Accuracy 81%
Now let's do the Neural Network
For that im gonna use the Relu activation function

Here are the results ,
The confusion matrix -
```
[[648 129]
 [159 264]]
```
Precision- 0.68 , Recall - 0.65, f1 score - 0.66

This shows us that , The Random Forest Classifier model gives us the better accuracy, this is due to the small amount of data we have , a neural network requires a large amount of data

With this Information lets plot the Power Corridor Infrastructure System Risk Priority Map



From this heatmap we can see that, there are some high risk areas that have scores greater than 0.8. There is no need for the pilots to waste drone battery flying along the standard route. Rather, flight routing can be optimized by targeting these dangerous points.

According to the grid positions, there are the following two sectors that have to be prioritized now:

Sector 1: Bottom Left – X: 0 to 10, Y: 50 to 60 (lots of dark spots).

Sector 2: Middle – X: 35 to 45, Y: 30 to 40

Making this end-to-end pipeline showed me how to do drone-based thermal imaging could be in detecting potential defects in electrical grids before an actual outage occurred. The most surprising thing i saw was that how the random forest model performed way better then the neural network , Despite the neural network sounding like a better approach, the Random Forest clearly outshined the MLP. After thinking about , i came to the logical conclusion that this disparity was due to having less data .

When considering future enhancements to this system, one of the limitations of this framework is the fact that a constant speed of a drone was used when projecting the 6,000 measurements to the simulated 100X60 as spatial_x and spatial_y grid. In reality, this process would greatly benefit from utilizing live GPS telemetry data. Another enhancement would be optimizing the machine learning model to take changes in ambient weather conditions into account . i would really like to improve the results by using more models like XGBoost , also some feature engineering is required .